

Caldwell B. Esselstyn, Jr., MD

# PREVENT AND REVERSE HEART DISEASE

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## Updating a 12-Year Experience With Arrest and Reversal Therapy for Coronary Heart Disease (An Overdue Requiem for Palliative Cardiology)

by Caldwell B. Esselstyn, Jr., MD

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Modern cardiology has given up on curing heart disease. Its aggressive interventions-- coronary artery bypass graft, atherectomy, angioplasty, and stenting--do not reduce the frequency of new heart attacks or prolong survival except in small subsets of patients.<sup>1'2</sup> Thus, it is clear that the goal of cardiology has become the relief of pain and unpleasant symptoms in the face of progressive disability and often death from disease. It is time to call this approach by its true name: palliative cardiology. It is also time to acknowledge that this approach is not the only alternative for our patients.

In this article, I will present converging lines of evidence (many of them well-known and universally accepted) reiterating that when serum cholesterol levels are maintained <150 mg/dl, coronary artery disease is practically nonexistent.<sup>3'4</sup> In a small group of my own patients, a 12-year follow-up shows prospective data confirming that a low-fat diet and lipid- lowering medication causes disease to halt or regress. I will also show that this fact can--and must--be made on the basis of a truly curative cardiology that prevents, halts, and selectively reverses heart disease.

Although coronary artery disease is the leading killer of men and women in the USA, it is rarely encountered in cultures that base their nutrition primarily on grains, legumes, lentils, vegetables, and fruit.<sup>3'4</sup> In the Framingham study, people with cholesterol levels between 150 and 200 mg/dl accounted for 35% of those with coronary heart disease, but among those with levels <150 mg/dl, the disease was rarely encountered.<sup>5'6</sup>

For over a decade it has been known that sufficient reduction of lipids may arrest and, in some cases, reverse coronary artery disease.<sup>7</sup> An analysis of 35 cholesterol-lowering studies confirms that the benefits are directly related to the degree of cholesterol reduction.<sup>8</sup>

This was dramatically confirmed when the Air Force/Texas Coronary Atherosclerosis Prevention Study of cholesterol reduction in >6,500 persons had to be prematurely stopped because of high mortality among controls, who had an average total cholesterol of 221 mg/dl.<sup>9</sup> During follow-up (mean 5.2 years), there were 95 new myocardial infarctions in the control group and 57 in the treated group.

Modern cardiology identifies patients with coronary heart disease through history, physical examination, and stress studies. Coronary angiography is usually performed. Patients with >70% diameter stenosis often receive aggressive, invasive interventions, including coronary artery bypass grafts, atherectomy, angioplasty, or stenting. Radiation may be added to decrease restenosis after angioplasty, and drugs are prescribed to decrease clotting. These are some of the reasons why the USA spends over a quarter of a trillion dollars a year on heart disease.<sup>10</sup> (In

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contrast, Canada, with fewer interventions, achieves equivalent survival rates in older patients with coronary artery disease.<sup>1)</sup>

Most patients who undergo these interventions do not have fewer new heart attacks or longer survival.<sup>2</sup> Life-threatening plaques are not directly treated.<sup>2</sup> The procedures themselves carry risks of new heart attacks, strokes, infections, encephalopathy, and mortality.<sup>2</sup> In addition, benefits erode with time. <sup>12</sup> A recent New England Journal of Medicine editorial pointed out that stents are overused and overpriced, and that some may be implanted without adequate anticoagulants, increasing thrombosis risk. <sup>13</sup> By using the mortality figures calculated from an earlier study,<sup>12</sup> 1.1% of the 1 million angioplasty, stenting, and atherectomy procedures performed every year and 1.3% of the 1 million annual coronary artery bypass graft procedures, we compute 240,000 deaths in a decade from these procedures.

These mechanical interventions treat only the symptoms, not the disease. It is therefore not surprising that patients who receive these interventions often experience progressive disease, graft shutdown, restenosis, more procedures, progressive disability, and death from disease. Thus, the leading killer of men and women in Western civilization is being left untreated. What is being practiced is "palliative cardiology": nontreatment of heart disease leading to disease extension and frequently an eventual fatal outcome.

In contrast, a 5-year experience has shown excellent results in patients with severe coronary artery disease who followed a plant-based diet containing <10% fat and who took cholesterol-lowering medication. <sup>14</sup> During this arrest and reversal therapy their lipid levels fell significantly, they experienced no new coronary events, and angiography showed that their disease had stabilized and in some cases selectively reversed. The goal at study onset in 1985 had been to achieve a total serum cholesterol of <150 mg/dl, the level seen in cultures where coronary artery disease is virtually absent.<sup>3</sup>

Today, after 12 years, I have followed the original patient cohort to determine adherence, safety, adverse effects, and long-term benefits. The original cohort contained 1 woman and 23 men, all nonsmoking, nondiabetic, and nonhypertensive patients with severe, angiographically demonstrated coronary artery disease. They agreed to follow a plant-based diet with <10% of calories derived from fat. They were asked to eliminate oil, dairy products (except skim milk and no-fat yogurt), fish, fowl, and meat. They were encouraged to eat grains, legumes, lentils, vegetables, and fruit. Cholesterol-lowering medication was individualized. The only goal was to achieve and maintain a total serum cholesterol of <150 mg/dl.

Six nonadherent patients were released within the first 12 to 18 months of the study, and they returned to standard care. By 1998, these patients, who initially had levels of angiographic and clinical disease equivalent to those of the adherent patients, had sustained 13 new cardiac events. The remaining 18 patients adhered to the study diet and medication for 5 years. At 5 years, 11 of these patients underwent angiographic analysis by the percent stenosis method, which demonstrated disease arrest in all 11 (100%) and regression in 8 (73%).<sup>14</sup> One patient admitted to the study with <20% left ventricular output died from a ventricular arrhythmia just weeks after the 5-year follow-up angiogram had confirmed disease regression. Autopsy revealed no myocardial infarction. Angina initially reported in 9 patients was eliminated in 2 and improved in the remaining 7. The patients' mean prestudy total cholesterol decreased from 237 to 137 mg/dl over 5 years.

During the 7 years since the conclusion of the 5-year study, all but 1 patient have continued to adhere to the prescribed diet and medication. Today, 12 years after study inception, the mean total cholesterol of the patients is 145 mg/dl (Table I).

**TABLE I** -- Lipid Profiles of 18 Patients Following a Very Low Fat Diet

| Patient | At Five Years             |             |             |                       | At 12 Years               |             |             |                       |
|---------|---------------------------|-------------|-------------|-----------------------|---------------------------|-------------|-------------|-----------------------|
|         | Total Cholesterol (mg/dl) | LDL (mg/dl) | HDL (mg/dl) | Triglycerides (mg/dl) | Total Cholesterol (mg/dl) | LDL (mg/dl) | HDL (mg/dl) | Triglycerides (mg/dl) |
| 1       | 137                       | 91          | 28          | 108                   | 138                       | 85          | 30          | 128                   |
| 2       | 136                       | 82          | 33          | 167                   | 138                       | 73          | 33          | 175                   |
| 3       | 130                       | 72          | 37          | 127                   | 145                       | 78          | 37          | 143                   |
| 4       | 124                       | 58          | 33          | 206                   | Deceased                  | -           | -           | -                     |
| 5       | 110                       | 65          | 29          | 97                    | 112                       | 64          | 30          | 96                    |
| 6       | 142                       | 75          | 52          | 78                    | 148                       | 80          | 41          | 109                   |
| 7       | 140                       | 77          | 43          | 117                   | 154                       | 86          | 40          | 155                   |
| 8       | 150                       | 63          | 32          | 362                   | 161                       | 77          | 30          | 351                   |
| 9       | 146                       | 90          | 40          | 91                    | 160                       | 89          | 41          | 93                    |
| 10      | 130                       | 76          | 41          | 72                    | 145                       | 90          | 44          | 99                    |
| 11      | 112                       | 48          | 31          | 188                   | 114                       | 48          | 33          | 180                   |
| 12      | 137                       | 92          | 26          | 113                   | 140                       | 90          | 27          | 133                   |
| 13      | 168                       | 85          | 64          | 78                    | 162                       | 84          | 68          | 65                    |
| 14      | 170                       | 98          | 38          | 254                   | 159                       | 87          | 45          | 183                   |
| 15      | 124                       | 72          | 32          | 104                   | 139                       | 85          | 35          | 119                   |
| 16      | 129                       | 62          | 26          | 226                   | 145                       | 70          | 27          | 246                   |
| 17      | 137                       | 81          | 39          | 68                    | 161                       | 100         | 48          | 60                    |
| 18      | 143                       | 88          | 35          | 113                   | 151                       | 102         | 29          | 103                   |
| Mean    | 137                       | 76          | 39          | 143                   | 145                       | 82          | 38          | 143                   |

Adherent patients have experienced no extension of clinical disease, no coronary events, and no interventions. This finding is all the more compelling when we consider that the original compliant 18 participants experienced 49 coronary events in the 8 years before the study.

These results are particularly important because they show that arrest and reversal therapy stops, rather than slows, coronary atherosclerosis. I argue that we must redefine what we mean by the phrase "the treatment of coronary atherosclerosis." We must shift the paradigm from interventional palliative cardiology to arrest and reversal therapy, which achieves numerical lipid goals that, when maintained, will abolish disease extension.

Adherence is the key factor upon which arrest and reversal therapy depends. Four techniques were used to promote adherence and reinforce the plant-based diet: (1) At enrollment, treatment objectives were discussed in an in-depth, 60- to 90-minute interview with each participant and spouse. (2) Patient adherence and lipid results were monitored through biweekly visits for the first 5 years. Such immediate recognition in achieving lipid goals is critical reinforcement and provides the patient with real-time proof of success. Visits became monthly during the second 5-year period and have been quarterly for the past 2 years. (3) Throughout the first year I called each patient on the evening of the clinic visit to review the lipid profile and any needed dietary or medication adjustments. (4) Several times a year a group meeting was held at my home or the home of a participant to review treatment objectives, exchange menus, and socialize. Continued frequent patient encounters appear critical to teach dietary knowledge and reinforce new habits. Patients reported that their physician's commitment to the same diet was additional motivation.

The study focused solely on lipid reduction through medication and diet, addressing, as Roberts has stated, "the only true risk factor for coronary artery disease--a cholesterol above 150 mg/dl."<sup>5</sup>

Combining a plant-based diet with medication achieves better long-term results than changing diet alone or combining modest diet changes and medication. 16-18 With such compelling long-term benefits, patients become empowered because they feel in control of the disease that was formerly destroying their lives.

The successful results from arrest and reversal therapy in this group of patients suggests it should be offered to all patients with coronary heart disease. These significant lipid reductions were undoubtedly because of our unrelenting persistence in dietary compliance combined with a statin

agent. Nevertheless, these reduced lipid levels are still in the range of normal for nations where the disease is absent.

Despite the benefits of a low-fat diet and of low lipid levels, the American Heart Association, the National Research Council, and the National Cholesterol Education Program recommend a 30% threshold for fat calories in the diet and a total cholesterol not >200 mg/dl.<sup>19'20</sup> But coronary artery disease develops and progresses with these guidelines, condemning millions of Americans to this epidemic.<sup>9'16.18</sup>

By way of contrast, no one maintaining a total serum cholesterol <150 mg/dl has succumbed to coronary artery disease in the Framingham study.<sup>6</sup> Campbell et al,<sup>4</sup> in the Cornell-China study, reports hundreds of thousands of rural Chinese going years without a single coronary event.

**Epidemiologic and evidence-based research has identified a lipid threshold for preventing the coronary artery disease epidemic. We have demonstrated that this threshold can be achieved and maintained with a plant-based diet and lipid-lowering medication when indicated, and that maintaining low lipid levels arrests and often reverses coronary artery disease.**

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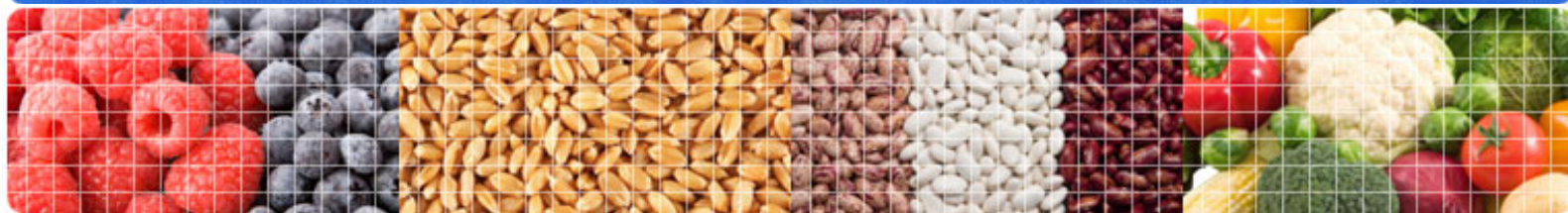
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## A SYNOPSIS OF "UPDATING A 12-YEAR EXPERIENCE WITH ARREST AND REVERSAL THERAPY FOR CORONARY HEART DISEASE (AN OVERDUE REQUIEM FOR PALLIATIVE CARDIOLOGY)"

by Caldwell B. Esselstyn, Jr., MD

Is there anything unusual, different, or profound about the present study? Unequivocally, yes!

Every patient with diagnosed coronary artery disease has the potential to make themselves heart attack proof and abolish any future disease progression by achieving and maintaining every 4-6 weeks a total serum cholesterol less than 150 mg/dL through ingesting a totally plant-based diet and cholesterol-lowering medication. Most of the patients in this study have been in the program for 14 years, making this experience the longest of its kind to date. These patients cherish the idea that they are the locus of control to eliminate the disease that was formerly destroying their lives. All compliant patients avoided any additional coronary events. The cholesterol reductions of these patients were greater than any comparable study to date - yet normal for cultures where the disease is virtually non-existent.

Most compelling about this study is that it provides every patient with a precise numerical threshold which, when regularly maintained, virtually assures them of freedom from disease. This is completely different from the present vague messages to patients who are left petrified as to when the other shoe might fall with disease extension. On the contrary, this method empowers patients to make themselves heart attack proof with a method proven through evidence-based research that will endure and continue to improve.

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